Journal of Autism and Developmental Disorders (2026) 56:1707–1724
https://doi.org/10.1007/s10803-024-06657-z

ORIGINAL ARTICLE

A Systematic Review of Implicit Versus Explicit Social Skills Group Programs in Different Settings for School-Aged Autistic Children and Adolescents

Bahareh Afsharnejad[1,4] · Patrice Whitehorne Smith[1,2] · Sven Bölte[1,2,3] · Ben Milbourn[1,2] · Sonya Girdler[1,2,3]

Accepted: 13 November 2024 / Published online: 17 December 2024
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024

Abstract

This systematic review and meta-analysis aimed to identify and evaluate the quality of randomised controlled trials (RCTs), assessing the efficacy of Social Skills Group Programs (SSGPs) for primary and secondary school aged autistic children and adolescents exploring the influence of informant, setting characteristics and teaching strategies as moderating factors for social outcomes. A search of the electronic databases of Medline, ProQuest, CINAHL, Scopus, and Web of Sciences electronic databases was conducted for the period January 1, 2013 until December 31, 2023 for peer-reviewed RCT studies published in English, evaluating the efficacy of SSGPs for school-aged autistic children and adolescents (6–18 years). Included studies were assessed for methodological quality and efficacy using random effect meta-analysis. Informant analysis and moderator analyses were also conducted investigating the influence of informant, setting characteristics and teaching strategy utilised in the SSGPs on the participants social outcomes. Sixty-five studies were included in the systematic review with 49 studies included in the meta-analysis. Although most studies had strong or good methodological quality, a high possibility of publication bias was detected in the meta-analysis. After statistical adjustments for publication bias were made, findings revealed that SSGPs had no effect on the overall outcomes assessed by included studies on school-aged autistic children and adolescents. However, informant analysis revealed small effects reported by self-report, parent-proxy and researchers. Teaching strategies and setting characteristics were not significant moderators for the efficacy of SSGPs on the social outcomes of autistic children and adolescents. This review highlights the need for improvements in measurement frameworks for assessing social skills in autistic children and adolescents.

Keywords Autism · Students · Social skills group · Randomised controlled trial · Systematic review · Meta-analysis

Introduction

Autistic children and adolescents often experience differences in their social cognition and interactions compared to their neurotypical peers (Crompton et al., 2020). This includes their understanding and expression of verbal and non-verbal cues, recognizing the emotional states of themselves and others, and grasping the social norms governing the neurotypical social world (Davis et al., 2021; Morrison et al., 2019). These differences can make it difficult for autistic children and adolescents to understand and respond to social situations according to neurotypical norms (Crompton et al., 2020), negatively impacting their social interactions with their neurotypical peers and their ability to make and maintain friendships (Morris et al., 2021; Petrina et al., 2014). Social Skills Group Programs (SSGPs) have emerged as an effective dimension of interventional strategies to aid

✉ Bahareh Afsharnejad
Bahareh.afsharnejad@curtin.edu.au

1 Curtin Autism Research Group (CARG), Curtin University, Perth, WA, Australia

2 Curtin School of Allied Curtin Autism Research Group (CARG), Curtin University, Perth, WA, Australia

3 Centre of Neurodevelopmental Disorders (KIND), Centre for Psychiatry Research, Division of Neuropsychiatry, Department of Women's and Children's Health, Karolinska Institutet & Child and Adolescent Psychiatry, Stockholm Health Care Services, Stockholm County Council, Stockholm, Sweden

4 Curtin University, The School of Population Health, Perth, Australia

autistic children and adolescents in navigating and adapting to social situations as their neurotypical peers, with the goal of improving their social experiences and outcomes (Hotton & Cole, 2015; Miller et al., 2014). These programs are generally delivered by 2–3 facilitators to a group of 4–8 autistic participants with average to above-average intellectual ability (IQ > 70). Generally, SSGPs focus on varying aspects of social cognition, social knowledge and awareness, social communication, and performance (Wolstencroft et al., 2018). The efficacy of SSGPs in achieving their aim has been studied extensively, utilising various study designs (experimental/non-experimental), intensity (hours the participant is exposed to the program), and measurement framework. This has led to a variety of systematic review and meta-analyses of such studies being published in the past decade for autistic children and adolescents (Afsharnejad et al., 2023; Gates et al., 2017; Naveed et al., 2019; Soares et al., 2021; Wolstencroft et al., 2018).

Findings from a systematic review and meta-analyses of studies assessing the efficacy of SSGPs evaluated via a randomised controlled trial design (RCT) indicate a medium effect for improving autistic children and adolescents' navigation of a neurotypical social world (Afsharnejad et al., 2023; Gates et al., 2017) and improving social awareness, social motivation, social communication, and social cognition (Wolstencroft et al., 2018). These reviews also caution readers when interpreting and generalising these findings. Generally, the included studies had medium to low program fidelity (Afsharnejad et al., 2023), indicating that the extent to which they delivered the SSGP as initially planned was unclear. Further, the measurement frameworks employed in these studies utilised a variety of outcomes, largely focusing on measuring social knowledge rather than actual social performance (Wolstencroft et al., 2018), generally relying of data obtained from informants other than autistic children and adolescents themselves (Gates et al., 2017). Despite contributing to the current evidence on SSGPs, these reviews did not consider that SSGPs use different teaching strategies and techniques to reinforce social skills (Kaat & Lecavalier, 2014). Teaching strategies involve the methods or approaches used to facilitate the development or reinforcement of social skills in these programs. These strategies can be broadly divided into either explicit (e.g., didactic), implicit (performance-based), or a mixture of both (Kaat & Lecavalier, 2014). Explicit teaching strategies involve direct instruction from a therapist or facilitator using didactic techniques such as modelling, demonstrations, and scripted role-play scenarios. These strategies aim to teach new skills or reinforce previously learned skills with the facilitators engaging participants in structured learning activities and providing feedback on their performance (Guivarch et al., 2017; McMahon et al., 2013). Implicit teaching strategies, however, utilise games, interest or leisure-based activities to engage participants in social interactions. These strategies do not involve direct instruction in specific social skills; instead, they focus on independent observation and learning through engagement and naturalistic organic interactions, free play, and in-vivo practice (Guivarch et al., 2017; McMahon et al., 2013). In recognising the strengths and limitations of both approaches, some SSGPs combine these strategies, blending both explicit and implicit approaches (Radley et al., 2017). Additionally, SSGPs vary in relation their delivery context, being commonly delivered in clinical, school, or leisure settings. Existing reports of SSGP efficacy are mainly based on programs designed for clinical settings (Afsharnejad et al., 2023; Gates et al., 2017; Wolstencroft et al., 2018). These programs are generally held after school hours at locations often far from the children and adolescents' residential addresses or schools (Dean & Chang, 2021). As such, the opportunities for further practicing and generalizing learned skills to authentic social environments, like school, may not be possible or hard to achieve. Despite these limitations, the influence of teaching strategies (i.e. explicit, implicit or both) and setting characteristics (i.e., school, clinic or home environments) on SSGPs outcomes for autistic children and adolescents remains unknown. These factors likely play an important moderating role in the quality, efficacy and outcomes of autistic participants. Consequently, the current systematic review and meta-analysis aimed to address this gap by identifying and evaluating the quality of randomised controlled trials (RCTs) assessing the efficacy of SSGPs for school-aged autistic children and adolescents, exploring the influence of teaching strategies, setting characteristics, and informants as moderating factors for social outcomes.

Methods

This systematic review and meta-analysis was registered with PROSPERO (identifier CRD42021275852). The Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement guided the methodological procedures (Liberati et al., 2009).

Search Strategy and Eligibility Criteria

Five databases (Medline, CINAHL, Scopus, ProQuest, and Web of Sciences) were searched for scholarly peer-reviewed RCT studies published in English between January 1, 2013, and December 31, 2022, evaluating the efficacy of SSGPs for autistic school-aged children and adolescents (6–18 years). Each database was searched at the title, abstract, and keyword for six search term groups including (1) autism ("Pervasive developmental disorder", autis*, Asperger*, ASD, PDD, "high-functioning autis*", HFA),

(2) Program (program, treatment, training, therapy, intervention), (3) RCT ("randomised control*", "randomised control*", random*, RCT), (4) age group (teen*, adolescen*, youth, Juvenile, Child*, kids, preteen, "pre-adolesce*"), (5) social, and (6) group. All search terms were modified, exploded, and combined via Boolean operators to match the database being searched. Due to a delay in data analysis, the search was repeated using the same eligibility criteria for the period January 1, 2023 to May 31, 2024. Studies were included if they: (1) reported findings generated by randomised controlled trials or used an experimental design with randomised allocation to treatment or control conditions; (2) recruited individuals with a clinical diagnosis of autism; (3) recruited participant groups within primary and secondary school age; (4) reported social skills outcomes from SSGPs; and (5) published in English. Studies were excluded if they were not delivered in a group format, did not focus on the outcomes for autistic children and adolescents, or both.

Study Selection

The citations produced from the search were imported into the Endnote reference manager. After removing duplicates, the articles were divided into three batches. Three pairs of independent reviewers separately screened each batch against the eligibility criteria at the abstract, title, and full-text levels. Each pair discussed any discrepancies until consensus was achieved. The final articles included were checked by the two first authors (BA and PWS), who also checked the reference lists of retrieved studies to capture other relevant articles meeting the eligibility criteria.

Data Extraction

The Cochrane Handbook for Systematic Review Guidelines guided the development of this study's data extraction (Higgins et al., 2023) and collected data on participants' characteristics, study design, aim and target social skills, components of the intervention group, teaching strategies, description, structure and format of the program, setting characteristics, assessment time points, primary and secondary outcome measures, incentives, attrition rates methodological quality, data analysis approach and summary of results.

Assessment of Methodological Quality

The first Author, BA, held a workshop educating two research assistants in assessing methodological quality in accordance with the Standard Quality Assessment Criteria (Kmet et al., 2004). This 14-item checklist examined: (a) clarity of the aim and design; (b) sample size calculation; (c) control group selection; (d) randomisation process; (e) blinding to group allocation (participants, investigators collecting data, or both); (f) the robustness of outcome measures; (g) analytic methods including some estimates of variance; (h) the sufficiency of reported results; and, (i) relevance of the discussion. Each item is scored on a scale of 0 (not achieved) to 2 (criteria met). The total score is calculated by dividing the total raw score by the possible maximal score of the relevant items) and is then converted to percentage scores. Scores of over 80% indicated a strong, 70–80% an adequate, 50–10% good, and those below 50% a limited methodological quality (Lee et al., 2008). The two research assistants independently assessed the quality of the included articles, noting a 92% inter-rater reliability. They then discussed any discrepancies until reaching a consensus.

Meta-analysis Procedure

The meta-analytic procedures outlined by Liberati et al. (2009) were used to conduct nine main meta-analyses. Two investigated the effect of SSGPs on post-intervention outcome measures either at completion or 3-month follow-up. The remaining seven explored the effect post-completion of the SSGP on outcome measures across studies included in the meta-analysis-based outcome categories outlined below. Under each category, further meta-analyses were conducted to investigate informant influence on each of these categories on outcomes exclusively reported by self, parent, teacher, and researcher. The script and data required to replicate the process are available at https://osf.io/f6sm4/.

Term Parameters—Outcome Categories

To better facilitate the synthesis of the data and results, the outcome measures were grouped into seven categories by the first author (BA) to compensate for the variability and heterogeneity of the outcome measures used across studies in assessing the efficacy of SSGP (Table 1). The outcome categories were reviewed and agreed upon by all authors, resulting in seven categories: adaptive functioning, autistic-like traits/ autism characteristics, behavioural/emotional challenges, physiology, sensory, social outcomes and well-being. The adaptive functioning category included multidimensional measures that assessed autistic children and adolescents' ability to function, cope and navigate everyday life (McQuaid et al., 2021). Autistic-like traits/autism characteristics consisted of measures used in the diagnosis or quantifying of features of autism (American Psychiatric Association, 2022; Ridout & Eldevik, 2023). The behavioural/emotional challenges category included measures that evaluated both internalising and externalising behaviours, including emotional states and

Table 1 Categorisation of outcome measures

Category	Outcome measure
Adaptive Functioning	Adaptive Behaviour Assessment System II (ABAS), Adapted Skillstreaming Checklist (A-SC), the Diagnostic Analysis of Non-Verbal Behaviour (DANVA-2), Developmental Disabilities Children's Global Assessment (DD-CGAS), OSU Autism Clinical Global impression – Severity (CGI-S), Peabody Picture Vocabulary Test (PPVT), Systematic Analysis of Languag Transcripts (SALT), Vineland Adaptive Behaviour Scale (VABS)
Autistic traits/autism characteristics	Asperger Syndrome Diagnostic Scale (ASDS), Autism Behaviour Checklist (ABC), Autism Quotient (AQ), Autism Diagnostic Observation Schedule (ADOS), Adaptive Skillstreaming Checklist (ASC), Autism Treatment Evaluation Checklist (ATEC), Childhood Autism Rating Scale (CARS), Comprehensive Assessment of Spoken Language (CASL), Social Communication Questionnaire (SCQ), Social Responsiveness Scale (SRS)
Behavioural/ Emotional challenges	Aberrant Behaviour Checklist (ABC), Affection for Others Questionnaire (AOQ), Autism Behaviour Checklist (ABC), Autism Spectrum Rating Scale (ASRS), Behaviour Assessment System for Children (BASC), Behaviour Rating Inventory of Executive Functioning (BRIEF), Cambridge Mindreading Face-Voice Battery for Children (CAM-C), Child and Adolescent Symptom Inventory-4 ASD Anxiety scale (CASI-ANX), Child Behaviour Checklist (CBCL), Children Depression Inventory 2nd Edition (CDI-2), Children in Stress (CiS), Depressioninventar fur Kinder- und Jugendliche (DIKJ), Emotion Quotient (EQ), Emotion Recognition and Display Survey (ERDS), Experience Sampling Method (ESM), Eyberg Child Behaviour Inventory (ECBI), Emotional Lexicon Test (ELT), Levels of Emotional Awareness Scale for Children (LEAS-C), Multidimensional Anxiety Scale for Children 2nd Edition (MASC-2), Paediatric Anxiety Rating Scale, Mindful Attention and Awareness Scale (MAAS), (PARS) Perceived Stress Scale (PSS), Perth A-Loneness Scale (PALS), Revised Children's Manifest Anxiety Scale (RCMAS), the Spence Child Anxiety Scale (SCAS), Social Interaction Anxiety Scale (SIAS), Strength and Difficulties Questionnaire (SDQ), Test of Emotional Comprehension (TEC), the Trait Anxiety Inventory (TAI), Social Skills Improvement Scale (SSIS)
Physiology	Cortisol levels, Event Related Potential (ERP), Incidental Face Memory (IFM), Bruinnicks-Oseretsky Test of Motor Proficiency (BOT), Functional Magnetic Resonance Imagining (fMRI)
Sensory	Austim Behaviour Checklist (ABC- Sensory Behaviour subscale), Child Autism Rating Scale (CARS-Taste, smell, touch response subscale), Sensory Integration Praxis Tests (SIPT), Sensory Profile (SP), Sensory Profile School Companion (SPSC)
Social outcomes	Adolescent Social Behaviour Scale (ASBS), Circumplex Scale of Interpersonal Efficacy (CSIE), the Contextual Assessment of Social Skills (CASS), Conversation Rating Scale (CRS), Emotional Regulation and Social Skills Questionnaire (ERSSQ), Emotion Regulation Social Skills Questionnaire (ERSSQ), Efficacy of Social skills Training in Austim (ESTIA), Goal Attainment Scaling (GAS), the Friendship Survey (FS), Friendship Qualities Scale (FQS), Matson Social Skills Questionnaire (MSSQ), NEPSY-II, the Peer Interaction Paradigm (PIP), Quality of Play Questionnaire (QPQ), Quality of Socialisation Questionnaire (QSQ), Social Communication Questionnare (SCQ), Social Competence with Peers Questionnaire (SCPQ), Social Interaction Observation System, (SIOS), Social Motivation and Competence Scale (SMCS), Social Skills assessment Group Questionnaire (SSGQ), Social Skills Questionnaire (SSQ), Social Skills Improvement Scale (SSIS), Social Skills Rating System (SSRS), Theory of Mind Test (ToM_T), Theory of Mind-Advanced Test (ToM_AT), Theory of Mind_Strange Stories (SSToM), Theory of Mind Behaviour Checklist (ToM_BC), Test of Adolescent Social Skills Knowledge _ Revised (TASSK), Vineland Adaptive Behaviour Scale (VABS_Socialisation subscale), Walk in the Forrest
Well-Being	Health Related Quality of Life (HFQoL), Paediatric Quality of Life Inventory (Pedqol)

regulation (Beck et al., 2020). Social outcome measures assessed social skills such as social knowledge, social responsiveness, conversation skills, non-verbal communication skills and play skills (Moody et al., 2022). Physiology outcome measures assessed changes in body functions and hormone levels (Godoy et al., 2022), while sensory outcome measures evaluated changes in sensory processing and response (Dwyer et al., 2022). Well-being outcome measures assess health-related (physical and mental health) quality of life and overall functioning (Najeeb & Quadt, 2024).

Statistical Analyses

Several studies were excluded from the meta-analyses. Two sets of studies, Lopata et al., (2019, 2020) and Schohl et al. (2014) and Van Hecke et al. (2015), were excluded due to overlapping samples. Only the study with the more complete data set (Lopata et al., 2019; Schohl et al., 2014) were included in the meta-analysis (Gates et al., 2017). In case of incompatible or incomplete data set, authors of the studies were contacted to provide the relevant data. If no response was received the study was excluded from the meta-analysis. Some studies only reported medians or other data incompatible with the

meta-analytic procedures and were excluded from the meta-analysis. Two other studies were excluded that only reported on subscale scores, and one that only reported the long-term effects of an SSGP (6 months). These exclusions helped to decrease the heterogeneity across studies, rendering greater internal validity for the meta-analysis. Effect size estimates were calculated (F values or t values were used when means and standard deviations were not reported) with a bias correction using Hedges' g (Lakens, 2013). To facilitate a precise estimate of the studies' effect and to account for variances within and between studies, the effect sizes and variances within individual studies were aggregated (Borenstein et al., 2009) based on the seven outcome categories described above using RStudio Version 4.3.3 (RStudio Team, 2024) and its available packages (compute.es, and MAd; Del Re, 2013, 2015; Del Re & Hoyt, 2018). JASP application version 0.18.3 (https://jasp-stats.org) was used to perform the meta-analyses.

Due to the lack of availability of correlations based on categories between outcome measures, a coefficient value of 0.5 was established for each outcome category (Borenstein et al., 2009). A statistical significance of 0.05 was established for the analysis, with SSGP effect sizes (Hedges' g) specified as < 0.2 as small, 0.2–0.5 as medium and > 0.8 as a large effect (Lakens, 2013). Taus and Chi-Square statistics with inconsistency score (I2) of 25% indicating low, 50% indicating moderate and 75% indicating high/substantial levels of heterogeneity (Higgins et al., 2023). Baujat plots were used to assess the contribution of each study to heterogeneity (Kossmeier et al., 2020). If ten or more studies indicated high heterogeneity, a meta-regression moderator analysis was conducted. Further analysis was then performed to investigate the moderating effect of sample size, type of control group, methodological quality, context, age group (children/adolescents), sex (proportion of males), and social skills practice approach on the efficacy of the SSGP.

Publication *Bias*

Possible publication bias was estimated using the Funnel plots and Egger's test by plotting the observed effect size against standard errors on the y-axis (Egger et al., 1997). An additional sensitivity test was performed as a visual inspection of the funnel plots' asymmetry alone cannot account for publication bias (Bartoš et al., 2022). Hence, a robust bias correction was undertaken using JASP to adjust the findings if a significant publication bias at $\alpha = 0.01$ was detected in the results (Bartoš et al., 2022).

Results

Overall, the original and updated database searches yielded a total of 1691 references. After removing duplicate publications, 510 references were checked for eligibility criteria, resulting in 149 references. Upon reviewing the references in the articles, five further references were added to the list. A total of 89 studies were excluded due to not being a journal article (k = 3), not having an autism-only sample (k = 6), not using a randomised control trial design (k = 9), the program not being a social skills group program (k = 59), Only reporting on parent-related outcomes (k = 6), only abstract written in English (k = 1), and the sample not being between 6 to 18 years of age (k = 5). Out of the 65 articles included in the narrative synthesis, 16 studies were not included for meta-analysis, with two studies using the same sample, 12 not reporting pre- and post-data for the intervention and control group, and two only reporting on subscales. As such, 49 studies were included in the meta-analysis (Fig. 1).

Narrative Synthesis

There were 65 studies included, of which 50 were structured manualised programs utilising 34 different SSGPs. All except two studies indicated they were RCTs (k = 63, 97%), reporting that they employed experimental designs (Ireri, 2019; Pordanjani, 2021). All studies involved the random assignment of autistic children and adolescents to intervention or control conditions. In most cases, the control condition was treatment or service as usual and waitlist (k = 41, 63%). However, some studies (k = 24, 37%) involved active control conditions such as barn activities, arts and crafts, cooking groups and facilitated playgroups that provided an intervention-based comparison point for the targeted SSGPs. Close to half of the studies were conducted in the United States of America (k = 30, 46%), followed by Netherlands (k = 4, 6%), Australia (k = 4, 6%), Brazil (k = 2, 3%), Canada (k = 2, 3%), China (k = 2, 3%), Germany (k = 2, 3%), Italy (k = 2, 3%), Israel (k = 2, 3%), South Korea (k = 2, 3%), Sweden (k = 2, 3%), United Kingdom (k = 2, 3%), France (k = 1, 2%), Greece (k = 1, 2%), Kenya (k = 1, 2%), Iceland (k = 1, 2%), India (k = 1, 2%), Iran (k = 1, 2%), Poland (k = 1, 2%), Romania (k = 1, 2%), and Turkey (k = 1, 2%).

Methodological Quality

The methodological quality of the 65 included studies varied. Many of the studies provided an overview of the flow of participants in line with the CONSORT guidelines (Moher et al., 2010) (k = 46, 79%). Most studies (k = 43, 66%) had 'strong' methodological quality with scores ranging from 82 to 100%, other studies were deemed 'good' (k = 18, 27%), with scores ranging from 71 to 79% and a few studies were considered 'adequate' (k = 4, 6%) with scores ranging from 50 and 68%. The main methodological limitations of studies included: failure to explain methods in sufficient detail,

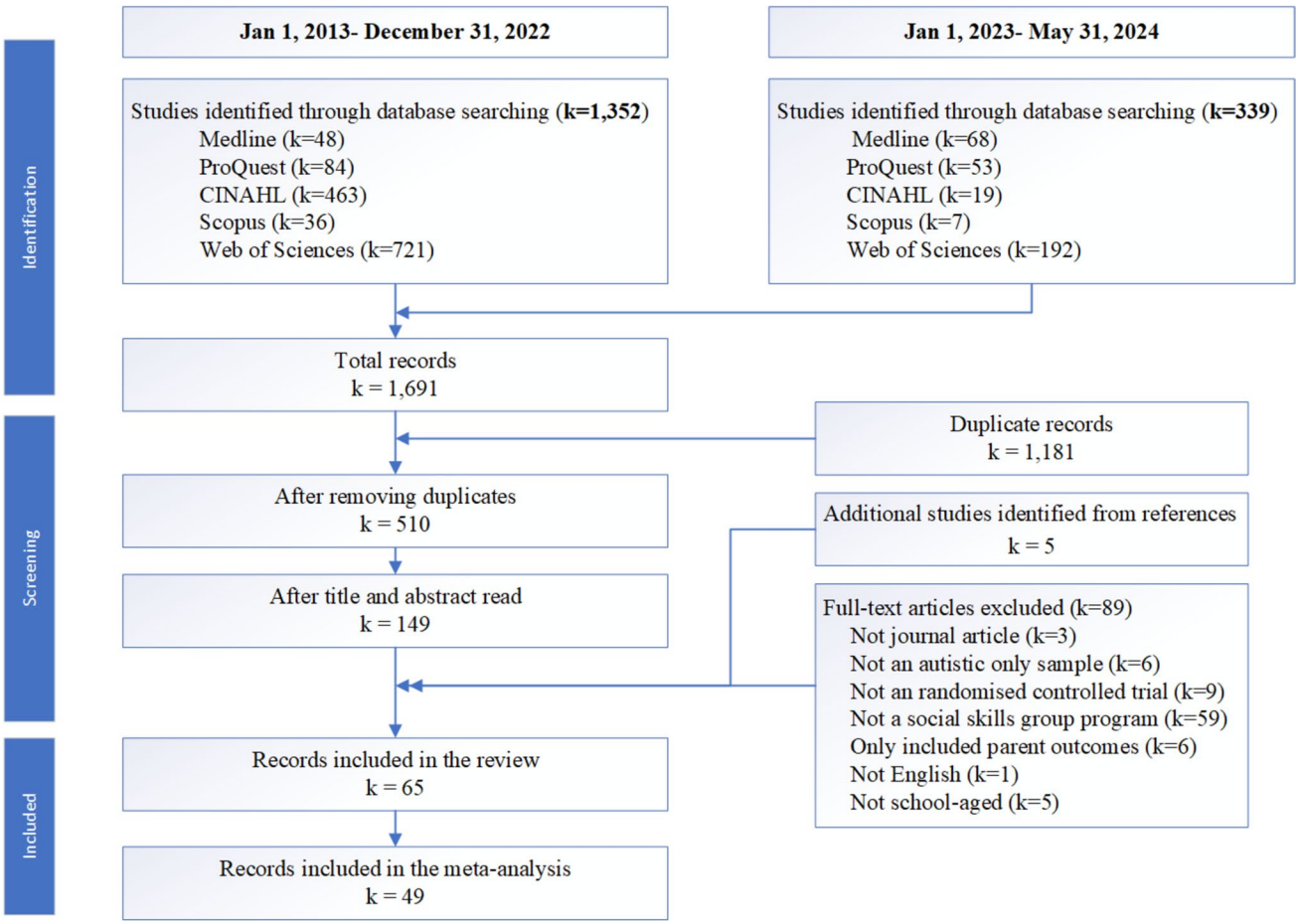


Fig. 1 The selection process of included studies in narrative review and meta-analyse of social skills group programs

failure to mask investigators, small sample sizes, failure to conduct intent to treat analysis and failure to report outcome measures in detail.

Participants and Participant Traits

There were 4554 autistic children and adolescents across the 65 included studies of which 3193 (70%) were identified as males and 1,361 (30%) as female. The age group of autistic participants spanned between 4 years old to 21 years old. Most of the included studies targeted children and adolescents who had been formally diagnosed, with an Intelligence Quotient (IQ) of 70 or higher (k = 37, 57%).

The majority of studies specified that autistic children and adolescents needed to have adequate verbal language skills, with no medical comorbidities (that would impact participation), serious mental health conditions or severe behavioural problems (k = 48, 74%).

Outcome Measures

All included studies evaluated the efficacy of interventions based on improvements in areas of functioning related to social cognition, social communication and social behaviour as primary outcomes. Other outcomes assessed in the included studies were related to adaptive functioning, autistic traits/autism characteristics, behavioural/emotional challenges, physiological and sensory functioning as well as well-being (Table 1). There were four types of informants used in the included studies to assess outcomes, namely parents, autistic children and adolescents, teachers and researcher/observer. The most common type of informants were parents of autistic children and adolescents (k = 58, 89%). The most common parent-report questionnaire utilised was the Social Responsiveness Scale (SRS) to evaluate children's social awareness, social cognition, social communication, social motivation, restricted interests and repetitive behaviour (k = 47, 72%).

Most studies also included direct assessments of autistic children and adolescents, and some included self-reports from autistic participants (k = 44, 68%). A wide range of assessment measures were administered to autistic children and adolescents to assess their outcomes. The tool most commonly reported was the Test of Adolescent Social Skills Knowledge (TASSK) (Laugeson et al., 2009) which evaluates social skills in adolescents who participated in the PEERS program (k = 10, 15%). Researchers/observers completed outcome measures in nearly half of the included studies (k = 29, 45%). There was a wide variation in the outcome measures used by research staff with the check (CASS/ DD-CGAS) Adapted Skillstreaming Checklist (A-SC) which assesses adaptive functioning of children diagnosed with autism without intellectual disability being the one most frequently used (k = 6, 9%). Teachers also provided assessment of autistic participants in some studies (k = 25, 38%), using the SRS. Overall, 29 (45%) collected follow-ups at a range spanning from 1 to 12 months. All these studies reported the changes observed post-completion of the program. As summary of all programs and the studies evaluating them is provided in Online Resource 1.

Setting Characteristics

Interventions were conducted in various settings. The most reported setting was clinics such as laboratories, child psychiatry/psychology clinics and outpatient centres (k = 28, 43%), followed by in school settings (k = 15, 23%) and leisure or recreational centres such as community centres, theatre and karate studies or riding centres (k = 12, 19%). Three interventions were conducted in the home environment (Beaumont, 2021; Fatta, 2023; Lee, 2023). However, seven studies did not report the setting of their intervention (k = 6, 11%). All but one SSGP (Costescu et al., 2017) conducted in the clinic setting indicated improvements in participants social skills (k = 27, 96%). Only the study conducted by Schwartzberg et al. (2015) in a leisure setting indicated mixed results related to improvement in social skills of participants, all other studies conducted in leisure settings indicated improvements in participants social skills (k = 11, 92%). All but two studies (Dickinson et al., 2016; Sansi et al., 2020) conducted in the school setting indicated gains in the social skills of participants (k = 13, 87%). All of the studies conducted in the home setting or that did not state a setting reported improvement in the social skills of participants (k = 10, 100%).

Social Skills Group Programs

The 38 SSGPs identified in the review varied in their aims, targeted social skills and components. The programs predominantly expressed using principles such as Parent involvement (k = 16; 42%), cognitive behaviour therapy (k = 13; 34%), neurotypical peer mediation (k = 10, 26%), psychoeducation (k = 5, 15%), theory of mind (k = 5, 13%), and observational learning (k = 5, 13%). Some programs expressed a broad goal of improving social skills or social functioning (k = 17; 45%). Others targeted a narrower goal, with emotion recognition/expression (k = 10, 26%) and interpersonal skills (k = 9, 24%) being the most favoured, followed by communication (k = 5, 13%) and self/emotion regulation (k = 5, 13%). The dosage of exposure to SSGPs ranged from once to five times weekly in 15- to 240-minute sessions spanning over one to 36 weeks. The programs were delivered to groups of 2 to 8 participants by various professions, including health professionals, trained facilitators, or experts in a specific activity (e.g., music, karate).

Teaching Strategies

The SSGPs assessed in the included studies utilised different teaching strategies, with 39% (k = 15) utilising only explicit strategies, 21% (k = 8) only implicit strategies and the rest (k = 15; 39%) using both. The most frequent strategy used across almost all programs utilising explicit approaches (k = 30) was direct teaching or instruction on how to perform social skills, recognise emotions, or problem-solve challenges associated with socialisation (k = 25, 83%). Role-play activities (k = 17; 57%), modelling by either the therapist or neurotypical peers (k = 13; 43%), homework (k = 12, 40%), practising skills with the group during a session (k = 11; 37%), and receiving feedback for performing a social skill (k = 10, 33%), were also favoured across programs. Strategies such as participants actively and independently brainstorming ideas through discussions and contributing to personally meaningful strategies to deliver a skill (k = 3; 11%) and mindfulness practices (k = 2; 7%) were rarely used. Amongst all SSGPs using implicit strategies (k = 23), unstructured socialisation through an activity such as music (k = 14; 61%) was the most utilised strategy, followed by social games (k = 10, 43%) and free play (k = 8, 35%).

Meta-analysis

Analysis of Outcomes from Baseline to Post-Test

A total of 205 effect sizes (*Mean* = 4.27 per study, *SD* = 2.80, *Median* = 4) from the 49 studies were included in this analysis. These studies overall had strong methodological quality (*M* = 0.82, *SD* = 0.09). In their RCT, 18 studies (27%)

employed active control groups, with the remaining being either usual care or waitlist controls. Eleven studies (22%) reported pooled results from children and adolescents. More than half of the studies (n = 27, 55%) utilised explicit strategies to practice social skills, eight (16%) used implicit strategies and the remaining employed a combination of both strategies (29%). Less than one-third of the programs were designed for delivery at school (n = 8; 16%) or a leisure context (n = 8; 16%).

The meta-analysis revealed an aggregated moderate overall effect for the efficacy of SSGPs for improving autistic adolescents' outcomes from baseline to post-test compared to the control group (Hedges' g = 0.49, $p = 0.003$, 95% CI [0.17, 0.82]. Due to the presence of publication bias, a Robust Bayesian meta-analysis adjustment was performed to account for the potential distortion of results. After adjusting the model, SSGPs no longer demonstrated efficacy in supporting autistic youth ($\mu = 0.04$; 95% CI [−0.59, 0.21]), further confirming the presence of publication bias (See Online Resource 2 for all forest plots).

Publication *Bias*

Publication bias was explored with Egger's regression test. Even in the absence of significant results on the Egger's regression test, funnel plots were inspected and a sensitivity analysis was performed to ensure a lack of publication bias. As demonstrated in Table 2, the majority of the analyses indicated effect sizes had high heterogeneity across studies, with publication bias also evident further pointing to the possible inflation of the reported evidence. Sensitivity analysis indicated that one study was significantly affecting the effect size (DeWalt et al., 2018). In such cases, statistical adjustments were conducted (Table 2).

Analysis of Maintenance Effects from Baseline to Follow-Up

The 17 studies included in the meta-analysis of maintenance effects resulted in effect sizes ranging from − 0.38 to 6.15, showing no significant effect ($p > 0.05$). This finding indicated that benefits reported post-completion of the program were not sustained to follow-up (Hedges' g = 0.70, $p = 0.06$, 95% CI [− 0.02, 1.41]). Egger's test and sensitivity analysis demonstrated no evidence for publication bias.

Analysis of Social Outcomes from Baseline to Post-Test

The meta-analysis of the 28 studies categorised under 'Social Outcomes' indicated effect sizes (Hedges' g) ranging from − 0.24 to 1.72. Findings from baseline to post-test indicated that despite observing a moderate effect, according to all informants, SSGPs were not significantly efficacious in improving the 'Social Outcomes' of autistic children and adolescents (Hedges' g = 0.74, $p = 0.14$; 95% CI [− 0.24, 1.72]). After adjusting the model due to the presence of publication bias, the reported effect for 'Social Outcomes' demonstrated a decrease from an unadjusted model to the adjusted one, suggesting small efficacy for SSGP these outcomes (μ = 0. 26, 95% CI [− 0.28, 0.96]).

Table 2 Meta analysis of key findings

Meta-analysis	Unadjusted ES	Egger's test (p-value)	τ^2	Presence of Publication bias	Adjusted ES
All outcomes from baseline to post-test	0.49**	< .001	1.34	Yes	0.04
All outcomes from baseline to follow-up	0.7	NS	2.25	No	
Social outcomes	0.74	NS	6.75	Yes	0.26
Self-report	0.6***	,01	0.2	Yes	0.46***
Parent Proxy-report	0.37***	NS	0.17	No	
Researcher	0.63***	NS	0.39	Yes	0.35***
Teacher Proxy-report	0.04				
Behavioural/Emotional challenges	0.54***	< .001	0.24	Yes	0.39*
Self-report	0.44***	< .001	0.22	Yes	0.46***
Parent Proxy-report	0.17				
Researcher	0.48**	NS	0.18	Yes	0.21**
Adaptive functioning	0.22				
Physiological Responses	0.44*	NS	0.16	Yes	0.25*
Autistic traits	0.45***	NS	0.41	Yes	0.30

Note: NS = Not significant. $I^2 = 100$ across all studies
*$P < .05$; **$P < .01$; ***$P < .001$

Child and Adolescent's Self-Reported Social Outcomes

The meta-analysis of the 17 studies, including child and adolescent's self-reported 'Social Outcomes', showed youth reporting moderate efficacy for SSGPs compared to control (Hedges' g = 0.60, $p < 0.001$; 95% CI [0.38, 0.83]). Adjusting the effect sizes suggested a modest efficacy for SSGP in supporting 'Social Outcomes' (μ = 0.46, 95% CI [0, 0.75]), a decrease from the unadjusted model.

Parent Proxy-Reported Social Outcomes

The meta-analysis of the 13 studies, including parent proxy-reported 'Social Outcomes', showed parents reporting small efficacy for SSGPs compared to control (Hedges' g = 0.37, $p < 0.001$; 95% CI [0.15, 0.59]). No publication bias was detected.

Researcher/*Assessor* Reported Social Outcomes

The meta-analysis of the 13 studies, including 'Social Outcomes' collected via a researcher/assessor, indicated a moderate efficacy for SSGPs compared to control (Hedges' g = 0.63, $p < 0.001$; 95% CI [0.28, 0.98]). Due to publication bias, the model was adjusted. The adjusted model highlighted a decrease in SSGP efficacy in supporting 'Social Outcomes' from the unadjusted model to the adjusted one, indicating small effects (μ = 0.35, 95% CI [− 0.15, 0.91]).

Teacher Proxy-Reported Social Outcomes

The meta-analysis of the 13 studies, including 'Social Outcomes' collected via teacher proxy reports, did not show a significant effect for the SSGP compared to the control group ($p > 0.05$).

Analysis of Behavioural/Emotional Challenges from Baseline to Post-Test

The meta-analysis of the 23 studies categorised under 'Behavioural/Emotional Challenges' indicated effect sizes (Hedges' g) ranging from 0.33 to 0.74. Findings showed that according to all informants, from baseline to post-test, SSGPs have been moderately efficacious in decreasing 'Behavioural/Emotional Challenges' of autistic children and adolescents (Hedges' g = 0.54, $p < 0.001$; 95% CI [− 0.33, 0.74]). The model was adjusted to account for the presence of publication bias. The adjusted model demonstrated a decreased effect (small) for SSGPs in reducing 'Behavioural/Emotional Challenges' compared to control groups than the unadjusted ones (μ = 0. 39, 95% CI [0, 0.61]).

Child and Adolescent's Self-Reported Behavioural/Emotional Challenges

The meta-analysis of the 15 studies, including self-reported 'Behavioural/Emotional Challenges', showed youth reporting moderate efficacy for SSGPs compared to control (Hedges' g = 0.44, $p < 0.001$; 95% CI [0.20, 0.68]). Adjusting the effect sizes due to publication bias suggested a modest efficacy for SSGPs in supporting 'Behavioural/Emotional Challenges' (μ = 0.40, 95% CI [0.25, 0.54]), a decrease from the unadjusted model.

Parent Proxy-Reported Behavioural/Emotional Challenges

The meta-analysis of the 12 studies, including parent proxy-reported 'Behavioural/Emotional Challenges', did not show parents reporting SSGP being efficacious in improving behavioural/emotional challenges of autistic children and adolescents compared to control (Hedges' g = 0.17, $p = 0.1$; 95% CI [−0.03, 0.37]). No evidence for publication bias was observed.

Researcher/*Assessor* Reported Behavioural/Emotional Challenges

The meta-analysis of the seven studies, including 'Behavioural/Emotional Challenges' collected via a researcher/assessor, indicated a moderate efficacy for SSGPs compared to control (Hedges' g = 0.48, $p = 0.003$; 95% CI [0.16, 0.80]). Due to the presence of publication bias, a model correction was applied. The adjusted model highlighted a decrease in SSGP efficacy for supporting 'Behavioural/Emotional Challenges' from the unadjusted model to the adjusted one, indicating small effects (μ = 0.21, 95% CI [0.12, 0.30]).

Analysis of Adaptive Functioning from Baseline to Post-Test

The meta-analysis of the seven studies categorised under the 'Adaptive Functioning' category indicated effect sizes (Hedges' g) ranging from − 0.27 to 0.75. Findings showed that according to all informants, from baseline to post-test, SSGPs have not been efficacious in improving the adaptive functioning skills of autistic children and adolescents (Hedges' g = 0.22, $p = 0.14$; 95% CI [− 0.07, 0.51]).

Analysis of Physiological Responses from Baseline to Post-Test

The meta-analysis of the five studies categorised under 'Physiological Responses' indicated effect sizes (Hedges' g) ranging from − 0.01 to 0.77. Findings showed that according to all informants, from baseline to post-test, SSGPs have been successful in moderately improving the physiological responses of autistic children and adolescents (Hedges' g = 0.44, p = 0.016; 95% CI [0.08, 0.79]). The model was adjusted due to publication bias. The adjusted model demonstrated a decreased effect (small) for SSGPs under this category compared to control groups than the unadjusted ones (μ = 0. 25, 95% CI [− 0.002, 0.74]).

Analysis of Autistic Traits from Baseline to Post-Test

The meta-analysis of the 30 studies categorised under 'Autistic Traits' (e.g. social awareness, communication, motivation, cognition) indicated effect sizes (Hedges' g) ranging from − 0.79 to 0.68. Findings showed that according to all informants, from baseline to post-test, SSGPs have moderately decreased Autistic Traits of autistic children and adolescents (Hedges' g = 0.45, $p < 0.001$; 95% CI [0.22, 0.68]). The model was again adjusted to account for the presence of publication bias. The adjusted model no longer demonstrated a significant effect in reducing 'Autistic Traits'.

Moderator Analysis

For all categories, except Adaptive Functioning and Physiological Responses, moderation analysis was conducted as the meta-analysis included over ten studies and enough variation across effect sizes (I^2) was observed. The meta-regression moderator analysis was significant only for 'Behavioural/Emotional Challenges', indicating the age group as a significant factor in pulling the results. Findings indicated that studies including both children and adolescents had a significantly lower effect compared to those only conducted with children (z = − 2.12, p = 0.03), while there was no statistically significant difference between the studies with only children or adolescents (z = − 0.86, p = 0.40).

Discussion

This systematic review and meta-analysis aimed to identify and evaluate the methodological quality of randomised controlled trials (RCTs) assessing the efficacy of SSGPs for primary and secondary school-aged autistic children and adolescents, exploring the influence of informant, setting characteristics and teaching/practice strategies as moderating factors for social outcomes. There were 65 studies included in the systematic review, with 49 of them assessed in the meta-analysis. Similar to previous reviews, although most studies demonstrated good to strong methodological quality (Afsharnejad et al., 2023; Gates et al., 2017), they were bounded by several limitations. Failure to mask investigators and participants (when possible) to intervention and control groups, as observed in many studies, may have increased the risk of expectancy bias (the tendency for expectations to influence perceptions and outcomes) and the placebo effect (believing the program has improved outcomes despite a lack of actual effect; Steeger et al., 2021). A large portion of the studies either had a very small sample size, failed to sample size calculation, or both. Studies with small sizes are more susceptible to inflated effect sizes and publication bias, decreasing confidence in the accuracy of their reported findings (Button et al., 2013). Finally, the inadequate description of the study design and randomisation across some studies, could contribute to a biased estimates of SSGP effect, affecting the rigour of a RCT, and threatening proper ethical standards (Altman et al., 2001).

Do social skills group programs improve outcomes?

Despite the narrative synthesis and previous literature indicating small to moderate overall effects for SSGPs in supporting autistic children and adolescents' social skills (Afsharnejad et al., 2023; Gates et al., 2017; Naveed et al., 2019; Soares et al., 2020; Wolstencroft et al., 2018), this meta-analysis revealed an adjusted null effect for the studies included in the meta-analysis. Notably, all these analyses were limited by either the inclusion of a selected group of outcome measures chosen based on the similarity (Naveed et al., 2019; Soares et al., 2020; Wolstencroft et al., 2018) or informant (Gates et al., 2017) or reporting outcomes for only adolescents (Afsharnejad et al., 2023), providing a limited picture of the efficacy of SSGPs. The variability of the SSGPs' principles and aims has led to a broad range of outcomes completed by varied informants being used across studies to assess their efficacy. The lack of a standardised measurement framework, even after controlling for the informant, can explain the limited overall efficacy observed from the meta-analyses.

Do assessment category results differ across informants?

While the meta-analysis of the 'Social Outcomes' did not indicate a significant effect for SSGPs, informant-based analysis of this category told a different story. The reported small effect related to 'Social Outcomes' by parents and researchers was comparable to the previous meta-analysis of studies based on informant reports, suggesting the success of such programs in supporting autistic children and adolescents' social knowledge and performance (Gates et al., 2017). However, unlike the previously reported large effect (Gates et al., 2017), the small to medium effects on the 'Social Outcomes' category, as reported by the autistic children and adolescents themselves, were more comparable

to those of their parents and researchers. This finding challenges the reported overestimation of social performance by autistic individuals compared to other informants (Lerner et al., 2012), attesting to the reliability of autistic participants self-reported changes and the importance of collecting this information (Thielmann et al., 2017). As this self-enhancement bias is comparable to those for a neurotypical population (Carlson & Kenny, 2012), a further investigation of the accuracy of the reported intensity of such biases is required.

Interestingly, similar to previous research, teachers reported no significant SSGP effect in this category (Gates et al., 2017). Across the studies collecting teacher reports, the context these teachers considered in completing questionnaires (e.g. class, recess, lunch break), the dosage of their daily contact and their level of familiarity with the participant are largely unknown. Teachers often report time constraints, associating it with their heavy daily workload of meeting their class's educational outcomes and overseeing a multitude of students with varying needs (Symes & Humphrey, 2010). This limited time availability can be the main driver of lower response rates noted in autism research (Al Jaffal, 2022). Also, sheds light on teachers' focus on the higher frequency of behavioural and emotional challenges of autistic students compared to their neurotypical counterpart (Ashburner et al., 2010) and their desire for learning strategies to better understand and manage such behaviours (Lindsay et al., 2013). Consequently, it is likely that the 'Social Outcomes' observed by teachers, particularly if they are less familiar with a student, are limited in scope and represent the frequency of overt social behaviours in the classroom (Messinger et al., 2019). Future studies employing teacher reports would benefit from asking teachers to allow time for observing the social competency and performance of a student across different contexts and activities.

Interestingly, the meta-analysis of the categories indicated the largest adjusted effect for 'Behavioural/Emotional Challenges', indicating the success of SSGPs in improving self and emotion regulation while decreasing mental health-related challenges. This is further supported by the decrease in the 'Physiological Responses' category. In the majority of SSGPs, this is not the main aim of the program. Yet, the depression and anxiety autistic people experience are mostly related to a lack of social connections and social isolation (Schiltz et al., 2022). As such, skills practised during the SSGP may have lessened the impact of mental health challenges as they aided autistic children and adolescents in navigating the complexities of neurotypical social interaction and relationship norms. Findings of the 'Behavioural/Emotional Challenges' meta-analysis, however, do not support the notion that SSGPs contribute to increased mental health concerns arising from masking and camouflaging (Cook et al., 2021). It has been suggested that SSGPs unintentionally stigmatise being autistic, implicitly or explicitly indicating the autistic individual needs to mask and act in a neurotypically accepted way. Possibly, these negative mental health outcomes are not observable in a typical 12-week follow-up period, being the prolonged effects of participating in SSGP. A paucity of collecting adverse events and investigating the longer effects these programs may have on the person is evident across the majority of the included studies, highlighting a need for further research in this field. Similar to 'Social Outcomes', the reported SSGP effects differed across informants, with parents observing no effects, researchers reporting small effects, and participants small to medium effects. Parents are often considered the main source in providing behavioural and emotional regulation support for their autistic children (Ting & Weiss, 2017), though the amount of time spent with them is limited to outside school hours. The shared time considerably decreases as children get older, with a decline in adolescent disclosure, especially emotional states (Main et al., 2019), to parents (Dubas & Gerris, et al., 2002), explaining the possible lack of significant effect reported by parents.

Can teaching strategy and context influence the effects of social skills group programs?

The availability of sufficient studies to run a moderator analysis on how the teaching strategy of SSGPs and their context influence the effects of SSGPs, has made this review unique in contributing to the greater body of knowledge. Despite the large variability in the teaching strategies and context across SSGPs, these factors did not influence the observed effects. This finding is supported by meta-analyses of non-specialist-mediated (Naveed et al., 2019) and technology-based programs (Soares et al., 2021) showing comparable findings to clinically-based, health professional-delivered SSGPs. Additionally, there is limited knowledge of how different components used in SSGPs accommodate increased social competence (Bottema-Beutel et al., 2018). Notably, findings did not indicate a moderating effect for teaching strategies on any assessment categories. Despite the knowledge of the large variability in autistic traits, social competency, and preferences across autistic individuals, research has failed to investigate whether these person-related factors can moderate the effects of SSGP based on their teaching strategies, indicating a need for future research in this field.

Interestingly, age emerged as a moderating factor for 'Behavioural/Emotional Challenges', indicating a higher effect for studies that included children compared to those that included both children and adolescents. This finding aligns with the developmental stage of Adolescence and its association with heightened emotional intensity and instability when responding to social cues (Casey et al.,

Table 3 Practical implications for future social skills group programs

Strategy	Explanation
Participatory approach to research	
Co-producing social skills group programs by implementing autistic voices in the development, assessment, and delivery of these programs	Many social skills group programs are developed based on the feedback of autistic participant; however, none are coproduced with the autism community to apply their preferences (Afsharnejad et al., 2023). The wide age range of participants—both developmentally and chronologically (Casey et al., 2019)—in SSGPs, combined with the absence of co-produced content tailored to the relevant consumer group, can hinder the effectiveness of narratives used in scripted activities and instructions. This limitation affects the pragmatic development of language and the quality of conversational exchanges. Accepting the diversity of the human brain under the neurodiversity paradigm necessitates that the voices, opinions, and values of the autistic community are incorporated into SSGPS (Davis & Crompton, 2021). Involving the autistic community in all stages of development, assessment, and delivery of SSGPs will ensure respect for their dignity, social intelligence, and self-advocacy
Practising Inclusivity	
Implementing Autistic-affirming approaches	The autistic advocacy groups have suggested that the sole reliance on explicit teaching strategies has been linked with a higher distortion of authenticity and lower positive self-appraisal, leading to camouflaging behaviours (Cook et al., 2021). Social skills group programs incorporating self-agency, such as brainstorming ideas with other autistic individuals to problem-solve social complexities, can promote a sense of autonomy, leading to better mental health outcomes
Redefining social skills-related concepts through a neurodiversity lens	Despite the recognition of autistic people outperforming their peers in some aspects, they are identified through the multitude of challenges they experience, suggesting a diagnosis of autism is linked to the absence of skills (Taylor et al., 2023). It would be beneficial for social skills group programs to incorporate the strengths of autistic people in accommodating their social functioning
Facilitating communication via alternative communication technology	The benefits of augmentative and alternative communication technologies for minimally verbal autistic individuals are well established (O'Donoghue et al., 2021). Recent literature suggests that regardless of verbal capabilities, facilitating communication technologies can support social engagement and communicative abilities across contexts (Zisk and Dalton, 2019)
Real-life practise by creating opportunities to practise learned skills in the person's daily social context	Despite schools being the most important social arena for students, the majority of evidence-based social skills group programs are delivered in contexts other than school, with no evidence of learned skills being generalised to school. Delivery of such programs at school can significantly improve everyday life social interactions, decrease anxiety, improve quality of life and foster a positive school environment (Fridell et al., 2023)
Providing a safe environment for practising social skills and graded exposure to a mixed group of autistic and neurotypical	Autistic individuals appreciate the safe space social skills group programs, including only autistic participants, provide for them to practise their skills (Crompton et al., 2020). They enjoy being able to connect and communicate with other autistics, expressing a feeling of belonging. Perhaps there is a need for graded exposure to a mixed group of neurotypicals and autistic people, with the number of neurotypicals increasing gradually

Table 3 (continued)

Strategy	Explanation
Enriching the social context	
Educating parents, increasing their awareness of neurodiversity and the strengths and support needs of their autistic children, to increase acceptance	Participant and parent psychoeducation and informing parents of ways they can support their child practice and generalise the learned skills have been the core elements of many studies (Afsharnejad et al., 2023). Such approaches focus mainly on the challenges associated with autism, perhaps leading to both participants and parents attributing the diagnosis as something negative, leading to experiencing negative self-appraisals. Educating parents about the diversity of the human brain and how some of their children's behaviour accommodates their needs rather than educating them on how to make them more neurotypical is a necessity
Developing, assessing and delivering programs that promote inclusivity at schools	According to the double empathy problem, misunderstandings between autistic and Neurotypical individuals arise from differences in how they view and experience the world (Milton & Green, 2024). Historically, this misunderstanding was solely attributed to the social deficits of autistic individuals, however, defining the double empathy problem acknowledges the reciprocal nature of the interaction, placing the responsibility on both parties (Milton & Green, 2024). This new knowledge highlights the importance of raising awareness and acceptance of autistic individuals' interaction styles to challenge the stigmatisation and discrimination associated with their diagnosis and create more inclusive environments that celebrate diversity (Finke and Dune, 2023)
Accounting for the diversity among participants	
Incorporating elements accounting for the specific needs of different genders	In line with the reported ratio of males to females, most evidence-based social skills group programs have been assessed with larger groups of males (Afsharnejad et al., 2023). Recent literature has indicated differences in the representation and social needs of females, and knowledge of other genders and their needs is mainly unknown. This limitation necessitates further research about the relevance of content to females and those identifying as non-binary
Accounting for the needs of those experiencing co-occurring conditions	The skills of autistic people and hence their needs are diverse, and perhaps they may require a combination of programs (Harkins & Mazurek, 2024). Social skills group programs content and strategy need to be considered in conjunction with co-occurring mental and physical health conditions, as they can play a large role in the participation of autistic youth; incorporating elements of psychoeducation about autism strengths and weaknesses along with its co-occurring conditions through a neuro affirming manner can be beneficial
Sustainability and quality of social skills group programs	
Sustainability and quality by ensuring programs are delivered as expected after assessment trials are completed	Evidence regarding the effect of social skills group programs is often collected by researchers delivering these programs, limiting the knowledge of intervention fidelity and sustainability after research is completed or when researchers are no longer present to oversee the delivery (Afsharnejad et al., 2023). As such, a clear protocol for training and supervising further staff for the delivery of these programs is necessary

2019). As such, SSGPs targeting this age group may benefit from implementing emotional regulation components in their content.

Publication bias and social skills group programs.

The evidence for the existence of publication bias was observed across most meta-analyses and was further confirmed by the smaller effects observed after adjusting the models. The presence of publication bias can be explained by different factors contributing to it. Not including all the identified articles in the systematic review in the meta-analysis could have been a contributing factor. Future research should consider reporting their data in a way that facilitates conducting a meta-analysis (Jonsson et al., 2016). Yet, no studies were excluded based on the

effect size of the reported data, eliminating the assumption of the publication bias observed being linked to studies with smaller effects being excluded (Gates et al., 2017). Some studies only reported outcomes with significant positive results. The absence of reports on negative or null results could have led to the observed inflated positive effects found in a meta-analysis (Kicinski et al., 2015; Murad et al., 2018). Gates et al. (2017) argued that new SSGPs are more prone to reporting in such a way and may be the reason behind the observed inflated effect, suggesting further research and replication of results for existing SSGPs.

The publication bias can also be related to the differences in sample sizes across studies, with small sample sizes being observed as a methodological concern similar to previous research (Gates et al., 2017). Having a small sample size is associated with an inflated effect, highlighting the necessity for future research evaluating SSGPs via an RCT methodology to calculate the required sample size and factor for possible attrition rates.

Social competence is multidimensional in nature, with dimensions covering performing prosocial behaviours, regulating emotions, adapting to the rules governing the immediate social context, perceived acceptance by others, and perceived success in social interactions (Gómez-Ortiz et al., 2017). This multidimensionality has led to variability in the aims of the SSGP, and the outcomes used to determine their efficacy, which may have been another factor leading to publication bias (Augusteijin et al., 2019; West et al., 2010). Although it is acknowledged that social skills are context-based, there is an absence of an existing widely applicable self-report measure of social skills that considers the different dimensions of social competence, that is easy to administer and validated across populations and settings (Gates et al., 2017; Wolstencroft et al., 2018). A clear indication of the dimensions of social competence a SSGP is addressing can lead to the development and use of outcome measures unanimously used across studies, hence increasing the comparability of findings in future research.

Implications

Table 3 outlines some of the recommendations emerging from the narrative review for those involved in delivering evidence-based SSGPs to autistic children and adolescents.

Limitations

Although providing new knowledge about the efficacy of SSGPs, this review was bound by a few limitations. The exclusion of 16 studies from the meta-analysis due to lack of data can significantly affect the effects reported, especially in the moderator analysis. A replication of the findings with these studies can provide a clearer picture of the effects of SSGP and the factors affecting them. Effects from some studies were for a wide age group including both children and adolescents, yet it has been suggested that SSGP is more beneficial for adolescents. Thus, such studies should consider reporting separate effects for children and adolescents as well as combined results, facilitating a clear moderator analysis for age. This information can also assist future facilitators in choosing the SSGP based on its success for a certain age group. Due to the variability of outcomes used across studies, they were grouped into seven key categories, which could have limited clarity of meta-analysis findings. Notably, compared to SSGPs developed for clinical delivery, a small portion were delivered at schools, which may have affected moderator analysis findings related to context. Despite the availability of school-delivered SSGPs, due to cost, staffing, or training challenges, their efficacy is rarely assessed via rigorous methodological standards (Beaumont et al., 2019). More RCTs evaluating the efficacy of SSGPs at school can lead to improved knowledge of their efficacy and superiority to other contexts.

Conclusion

There has been an increase in RCTs assessing the efficacy of SSGPs on the social skills of autistic children and adolescents over the past decade. This systematic review and meta-analysis represent the largest review done to date in this area. Although SSGPs are the most used social skills interventions for children and adolescents, this review found an overall null effect for SSGPs considering outcomes from various categories, including social skills. The differences in the findings of this review compared to those previously conducted suggest the need for more rigorous assessment and reporting on social outcomes in SSGPs. It further indicates the need for SSGPs co-produced based with the autistic community, ensuring the relevance of the content to the needs of autistic community.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10803-024-06657-z.

Acknowledgements We wish to thank the Stan Perron's Charitable Foundation and the School of Special Education Needs-Disability for the funding received for this research. The funding institutions had no role in the study design, implementation, analysis, writing of the manuscript or decision to publish the paper. We also wish to thank Ms.

Diana Blackwood, Dr. Jim Parsons, and Ms. Sessina Figueiredo for their support with data extraction.

Author Contributions All authors contributed to the study conception and design. Data Analysis was performed by Bahareh Afsharnejad and Patrice Whitehorne-Smith. The first draft of the manuscript was written by Bahareh Afsharnejad and Patrice Whitehorne-Smith and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declarations

Conflict of interests There are no conflicts of interest in the publication of this article.

References

Afsharnejad, B., Black, M. H., Falkmer, M., Bölte, S., & Girdler, S. (2023). The methodological quality and intervention fidelity of randomised controlled trials evaluating social skills group programs in autistic adolescents: A systematic review and meta-analysis. *Journal of Autism and Developmental Disorders, 54*, 1281–1316. https://doi.org/10.1007/s10803-023-05893-z

Al, J. M. (2022). Barriers general education teachers face regarding the inclusion of students with autism. *Frontiers in Psychology, 13*, 873248. https://doi.org/10.3389/fpsyg.2022.873248

Altman, D. G., Schulz, K. F., Moher, D., Egger, M., Davidoff, F., Elbourne, D., ... & Lang, T. (2001). The revised CONSORT statement for reporting randomized trials: Explanation and elaboration. *Annals of Internal Medicine*, *134*(8), 663–694.

American Psychiatric Association (2022). Diagnostic and statistical manual of mental disorders, 5th ed., text rev.American Psychiatric Publishing: Washington, DC. https://doi.org/10.1176/appi.books.9780890425787

Ashburner, J., Ziviani, J., & Rodger, S. (2010). Surviving in the mainstream: Capacity of children with autism spectrum disorders to perform academically and regulate their emotions and behavior at school. *Research in Autism Spectrum Disorders, 4*(1), 18–27. https://doi.org/10.1016/j.rasd.2009.07.002

Augusteijn, H. E. M., van Aert, R. C. M., & van Assen, M. A. L. M. (2019). The effect of publication bias on the Q test and assessment of heterogeneity. *Psychological Methods, 24*(1), 116–134. https://doi.org/10.1037/met0000197

Bartoš, F., Maier, M., Quintana, D. S., & Wagenmakers, E. J. (2022). Adjusting for publication bias in JASP andR: Selection models, PET-PEESE, and robust Bayesian meta-analysis. *Advances in Methods and Practices inPsychological Science, 5*(3), 25152459221109259. https://doi.org/10.1177/25152459221109259

Beaumont, R. B., Smith-Merry, J., Costley, D., Howlin, P., Sofronoff, K., Roberts, J., ... & Einfeld, S. L. (2019). Implementation, evaluation and maintenance of a social-emotional skills training program for children with an autism spectrum disorder in a specialist school setting. *International journal of special education, 34*(1), 95–108.

Beaumont, R., Walker, H., Weiss, J., & Sofronoff, K. (2021). Randomised controlled trial of a video gaming-based social skills program for children on the autism spectrum. *Journal of Autism and Developmental Disorders, 51*(10), 3637–3650. https://doi.org/10.1007/s10803-020-04801-z

Beck, K. B., Conner, C. M., Breitenfeldt, K. E., Northrup, J. B., White, S. W., & Mazefsky, C. A. (2020). Assessment and treatment of emotion regulation impairment in autism spectrum disorder across the life span: Current state of the science and future directions. *Child and Adolescent Psychiatric Clinics of North America, 29*(3), 527–542. https://doi.org/10.1016/j.chc.2020.02.003

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). Multiple outcomes or time points within a study. In M. Borenstein, L. V. Hedges, J. P. T. Higgins & H.R. Rothstein (Eds.), *Introduction to Meta-Analysis* (pp. 225–238). John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470743386.ch24

Bottema-Beutel, K., Kapp, S. K., Lester, J. N., Sasson, N. J., & Hand, B. N. (2021). Avoiding ableist language: Suggestions for autism researchers. *Autism in Adulthood, 3*(1), 18–29. https://doi.org/10.1089/aut.2020.0014

Bottema-Beutel, K., Park, H., & Kim, S. Y. (2018). Commentary on social skills training curricula for individuals with ASD: Social interaction, authenticity, and stigma. *Journal of Autism and Developmental Disorders, 48*, 953–964. https://doi.org/10.1007/s10803-017-3400-1

Button, K. S., Ioannidis, J. P., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience, 14*(5), 365–376. https://doi.org/10.1038/nrn3475

Carlson, E. N., & Kenny, D. A. (2012). Meta-accuracy: Do we know how others see us? In S. Vazire & T. D. Wilson (Eds.), *Handbook of self-knowledge* (pp. 242–257). The Guilford Press.

Casey, B. J., Heller, A. S., Gee, D. G., & Cohen, A. O. (2019). Development of the emotional brain. *Neuroscience Letters, 693*, 29–34. https://doi.org/10.1016/j.neulet.2017.11.055

Constantino, J.N. & Gruber, C.P. (2012). *Social Responsiveness Scale-Second Edition* (SRS-2). Western Psychological Services

Cook, J., Hull, L., Crane, L., & Mandy, W. (2021). Camouflaging in autism: A systematic review. *Clinical Psychology Review, 89*, 102080. https://doi.org/10.1089/aut.2020.0071

*Costescu, C., Bram Vanderborght, & David, D. (2017). Robot-enhanced CBT for dysfunctional emotions in social situations for children with ASD. *Journal of Evidence-Based Psychotherapies*, *17*(2), 119–132. https://doi.org/10.24193/jebp.2017.2.7

Crompton, C. J., Hallett, S., Ropar, D., Flynn, E., & Fletcher-Watson, S. (2020). 'I never realised everybody felt as happy as I do when I am around autistic people': A thematic analysis of autistic adults' relationships with autistic and neurotypical friends and family. *Autism, 24*(6), 1438–1448. https://doi.org/10.1177/1362361320908976

Davis, R., & Crompton, C. J. (2021). What do new findings about social interaction in autistic adults mean for neurodevelopmental research? *Perspectives on Psychological Science, 16*(3), 649–653. https://doi.org/10.1177/1745691620958010

DaWalt, L. S., Greenberg, J. S., & Mailick, M. R. (2018). Transitioning Together: Multi-family group psychoeducation program for adolescents with asd and their parents. *Journal of Autism and Developmental Disorders, 48*(1), 251–263. https://doi.org/10.1007/s10803-017-3307-x

Dean, M., & Chang, Y. C. (2021). A systematic review of school-based social skills interventions and observed social outcomes for students with autism spectrum disorder in inclusive settings. *Autism, 25*(7), 1828–1843. https://doi.org/10.1177/13623613211012886

Dickinson, K., & Place, M. (2016). The impact of a computer-based activity program on the social functioning of children with autistic spectrum disorder. *Games for Health Journal, 5*(3), 209–215. https://doi.org/10.1089/g4h.2015.0063

Dubas, J. S., & Gerris, J. R. M. (2002). Longitudinal changes in the time parents spend in activities with their adolescent children as a function of child age, pubertal status and gender. *Journal of*

Family Psychology, 16(4), 415–427. https://doi.org/10.1037/0893-3200.16.4.415

Dwyer, P., Takarae, Y., Zadeh, I., Rivera, S. M., & Saron, C. D. (2022). A multidimensional investigation of sensory processing in autism: Parent-and self-report questionnaires, psychophysical thresholds, and event-related potentials in the auditory and somatosensory modalities. *Frontiers in Human Neuroscience, 16*, 811547. https://doi.org/10.3389/fnhum.2022.811547

Egger, M., Smith, G. D., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple,graphical test. *BMJ, 315*(7109), 629–634. https://doi.org/10.1136/bmj.315.7109.629

Fatta, L. A., Laugeson, E. A., Bianchi, D., Laghi, F., & Scattoni, M. L. (2024). Program for the education and enrichment of relational skills (PEERS®) for Italy: A randomised controlled trial of a social skills intervention for autistic adolescents. *Journal of Autism and Developmental Disorders*. https://doi.org/10.1007/s10803-023-06211-3

Finke, E. H., & Dunn, D. H. (2023). Neurodiversity and double empathy: can empathy disconnects be mitigated tosupport autistic belonging?. *Disability & Society,* 1–24. https://doi.org/10.1080/09687599.2023.2295802

Freitag, C. M., Jensen, K., Elsuni, L., Sachse, M., Herpertz-Dahlmann, B., Schulte-Rüther, M., ... & Cholemkery, H. (2016). Group-based cognitive behaviouralpsychotherapy for children and adolescents with ASD: The randomized, multicentre, controlled SOSTA–net trial.*Journal of Child Psychology and Psychiatry,57*(5), 596–605.https://doi.org/10.1111/jcpp.12509

Fridell, A., Coco, C., Borg, A., & Bölte, S. (2023). School-based social skills group training (SKOLKONTAKT™):A pilot randomized controlled trial. *Frontiers in Psychology, 14*, 1128288.https://doi.org/10.3389/fpsyg.2023.1128288

Gates, J. A., Kang, E., & Lerner, M. D. (2017). Efficacy of group social skills interventions for youth with autism spectrum disorder: A systematic review and meta-analysis. *Clinical Psychology Review, 52*, 164–181. https://doi.org/10.1016/j.cpr.2017.01.006

Godoy, P. B. G., Sumiya, F. M., Seda, L., & Shephard, E. (2022). A systematic review of observational, naturalistic, and neurophysiological outcome measures of nonpharmacological interventions for autism. *Brazilian Journal of Psychiatry, 44*, 532–547.https://www.scielo.br/j/rbp/a/PsWqcQWqdXQ8z7NxGHMdKPj/

Gómez-Ortiz, O., Romera-Felix, E. M., & Ortega-Ruiz, R. (2017). Multidimensionality of social competence: Measurement of the construct and its relationship with bullying roles. *Revista De Psicodidáctica (English Ed.), 22*(1), 37–44. https://doi.org/10.1387/RevPsicodidact.15702

Guivarch, J., Murdymootoo, V., Elissalde, S. N., Salle-Collemiche, X., Tardieu, S., Jouve, E., & Poinso, F. (2017). Impact of an implicit social skills training group in children with autism spectrum disorder without intellectual disability: A before-and-after study. *PLoS ONE, 12*(7), e0181159. https://doi.org/10.1371/journal.pone.0181159

Harkins, C., & Mazurek, M.O. (2024). The impact of co-occurring ADHD on social competence interventionoutcomes in youth with Autism spectrum disorder. *Journal of Autism and Developmental Disorders, 54*, 2552–2563. https://doi.org/10.1007/s10803-023-05987-8

Higgins, J. P. T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M. J., Welch, V. A., (Eds.) (2023). *Cochrane Handbook for Systematic Reviews of Interventions version 6.4.* www.training.cochrane.org/handbook.

Hotton, M., & Coles, S. (2016). The effectiveness of social skills training groups for Individuals with autism spectrum disorder. *Review Journal Autism Developmental Disorder, 3*, 68–81. https://doi.org/10.1007/s40489-015-0066-5

*Ireri, N. W., White, S. W., & Mbwayo, A. W. (2019). Treating Anxiety and Social Deficits in Children with Autism Spectrum Disorder in Two Schools in Nairobi, Kenya. *Journal of Autism and Developmental Disorders, 49*(8), 3309–3315. https://doi.org/10.1007/s10803-019-04045-6

Jonsson, U., Choque Olsson, N., & Bölte, S. (2016). Can findings from randomised controlled trials of social skills training in autism spectrum disorder be generalised? The neglected dimension of external validity. Autism: *The International Journal of Research and Practice, 20*(3), 295–305. https://doi.org/10.1177/1362361315583817

Kaat, A. J., & Lecavalier, L. (2014). Group-based social skills treatment: A methodological review. *Research in Autism Spectrum Disorders, 8*(1), 15–24. https://doi.org/10.1016/j.rasd.2013.10.007

Kicinski, M., Springate, D. A., & Kontopantelis, E. (2015). Publication bias in meta-analyses from the Cochrane Database of Systematic Reviews. *Statistics in Medicine, 34*(20), 2781–2793. https://doi.org/10.1002/sim.6525

Kmet, L. M. (2004). Standard quality assessment criteria for evaluating primary research papers from a variety offields. Alberta Heritage Foundation for Medical Research Edmonton.

Kossmeier, M., Tran, U. S., & Voracek, M. (2020). Charting the landscape of graphical displays for meta-analysis and systematic reviews: A comprehensive review, taxonomy, and feature analysis. *BMC Medical Research Methodology, 20*(1), 26. https://doi.org/10.1186/s12874-020-0911-9

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology, 4*, 62627. https://doi.org/10.3389/fpsyg.2013.00863

Laugeson, E. A., Frankel, F., Mogil, C., & Dillon, A. R. (2009). Parent-assisted social skills training to improvefriendships in teens with autism spectrum disorders. *Journal of Autism and Developmental Disorders, 39*, 596–606.https://doi.org/10.1007/s10803-008-0664-5

Lee, J. H., Lee, T. S., Yoo, S. Y., Lee, S. W., Jang, J. H., Choi, Y. J., & Park, Y. R. (2023). Metaverse-based social skills training programme for children with autism spectrum disorder to improve social interaction ability: An open-label, single-centre, randomised controlled pilot trial. *Eclinical Medicine, 61*, 102072. https://doi.org/10.1016/j.eclinm.2023.102072

Lerner, M. D., Calhoun, C. D., Mikami, A. Y., & De Los Reyes, A. (2012). Understanding parent-child social informant discrepancy in youth with high functioning autism spectrum disorders. *Journal of Autism and Developmental Disorders, 42*(12), 2680–2692. https://doi.org/10.1007/s10803-012-1525-9

Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P., Clarke, M., … & Moher, D.(2009) The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: Explanation and elaboration. *Annals of Internal Medicine, 151*(14), W–65.

Lindsay, S., Proulx, M., Thomson, N., & Scott, H. (2013). Educators' challenges of including children with autism spectrum disorder in mainstream classrooms. *International Journal of Disability, Development and Education, 60*(4), 347–362. https://doi.org/10.1080/1034912X.2013.846470

*Lopata, C., Thomeer, M. L., Rodgers, J. D., Donnelly, J. P., McDonald, C. A., Volker, M. A., Smith, T. H., & Wang, H. (2019). Cluster randomised trial of a school intervention for children with autism spectrum disorder. *Journal of Clinical Child and Adolescent Psychology: The official journal for the Society of Clinical Child and Adolescent Psychology, American Psychological Association, Division 53*, *48*(6), 922–933. https://doi.org/10.1080/15374416.2018.1520121

Lopata, C., Donnelly, J. P., Thomeer, M. L., Rodgers, J. D., Lodi-Smith, J., Booth, A. J., & Volker, M. A. (2020). Moderators of school intervention outcomes for children with autism spectrum disorder. *Journal of Abnormal Child Psychology, 48*(8), 1105–1114. https://doi.org/10.1007/s10802-020-00652-5

Main, A., Lougheed, J. P., Disla, J., & Kashi, S. (2019). Timing of adolescent emotional disclosures: The role of maternal emotions and adolescent age. *Emotion, 19*(5), 829–840. https://doi.org/10.1037/emo0000483

McMahon, C. M., Lerner, M. D., & Britton, N. (2013). Group-based social skills interventions for adolescents with higher-functioning autism spectrum disorder: A review and looking to the future. *Adolescent Health, Medicine and Therapeutics, 4*, 23–28. https://doi.org/10.2147/AHMT.S25402

McQuaid, G. A., Pelphrey, K. A., Bookheimer, S. Y., Dapretto, M., Webb, S. J., Bernier, R. A., McPartland, J. C., Van Horn, J. D., & Wallace, G. L. (2021). The gap between IQ and adaptive functioning in autism spectrum disorder: Disentangling diagnostic and sex differences. *Autism: The International Journal Of Research And Practice, 25*(6), 1565–1579. https://doi.org/10.1177/1362361321995620

Messinger, D. S., Prince, E. B., Zheng, M., Martin, K., Mitsven, S. G., Huang, S., Stölzel, T., Johnson, N., Rudolph, U., Perry, L. K., Laursen, B., & Song, C. (2019). Continuous measurement of dynamic classroom social interactions. *International Journal of Behavioral Development, 43*(3), 263–270. https://doi.org/10.1177/0165025418820708

Miller, A., Vernon, T., Wu, V., & Russo, K. (2014). Social skill group interventions for adolescents with autism spectrum disorders: A systematic review. *Review Journal of Autism and Developmental Disorders, 1*, 254–265. https://doi.org/10.1007/s40489-014-0017-6

Milton, D. E., & Green, J. (2024). Theorising autism. *Autism, 28*(4), 795–797.https://doi.org/10.1177/13623613241235786

Moher, D., Hopewell, S., Schulz, K. F., Montori, V., Gøtzsche, P. C., Devereaux, P. J., Elbourne, D., Egger, M., & Altman, D. G. (2010). CONSORT 2010 explanation and elaboration: Updated guidelines for reporting parallel group randomised trials. *BMJ (Clinical Research Ed.), 340*, c869. https://doi.org/10.1136/bmj.c869

Moody, C. T., Schiltz, H. K., Ratto, A. B., Gulsrud, A. C., & Laugeson, E. A. (2022). Measurement of social skills treatment outcome in autism: Moving beyond informant report and considering diversity. *International Review of Research in Developmental Disabilities 62, 265–312*. Academic Press. https://doi.org/10.1016/bs.irrdd.2022.05.007

Morris, S., O'Reilly, G., & Nayyar, J. (2021). Classroom-based peer interventions targeting autism ignorance, prejudice and/or discrimination: A systematic PRISMA review. *International Journal of Inclusive Education*, 1–45. XXXXhttps://doi.org/10.1080/13603116.2021.1900421

Morrison, K. E., Pinkham, A. E., Kelsven, S., Ludwig, K., Penn, D. L., & Sasson, N. J. (2019). Psychometric evaluation of social cognitive measures for adults with autism. *Autism Research: Official Journal of the International Society for Autism Research, 12*(5), 766–778. https://doi.org/10.1002/aur.2084

Murad, M. H., Chu, H., Lin, L., & Wang, Z. (2018). The effect of publication bias magnitude and direction on the certainty in evidence. *BMJ Evidence-Based Medicine, 23*(3), 84–86. https://doi.org/10.1136/bmjebm-2018-110891

Naveed, S., Waqas, A., Amray, A. N., Memon, R. I., Javed, N., Tahir, M. A., ... & Rahman, A. (2019). Implementation and effectiveness of non-specialist mediated interventions for children with autism spectrum disorder: A systematic review and meta-analysis. *PloS one*, *14*(11), e0224362. https://doi.org/10.1371/journal.pone.0224362

Najeeb, P., & Quadt, L. (2024). Autistic well-being: A scoping review of scientific studies from a neurodiversityaffirmative perspective. *Neurodiversity, 2*, 27546330241233088. https://doi.org/10.1177/27546330241233088

O'Donoghue, M., O'Dea, A., O'Leary, N., Kennedy, N., Forbes, J., & Murphy, C. A.(2021). Systematic review of peer-mediated intervention for children with autism whoare minimally verbal. Review *Journal of Autism and Developmental Disorders, 8*, 51–66.https://doi.org/10.1007/s40489-020-00201-2

O'Haire, M. E. (2013). Animal-assisted intervention for autism spectrum disorder: A systematic literature review. *Journal of Autism and Developmental Disorders, 43*(7), 1606–1622. https://doi.org/10.1007/s10803-012-1707-5

Petrina, N., Carter, M., & Stephenson, J. (2014). The nature of friendship in children with autism spectrum disorders: A systematic review. *Research in Autism Spectrum Disorders, 8*(2), 111–126. https://doi.org/10.1016/j.rasd.2013.10.016

Radley, K.C., O'Handley, R.D., Sabey, C.V. (2017). Social skills training for children and adolescents with autism spectrum disorder. In: Matson, J. (eds) Handbook of Treatments for Autism Spectrum Disorder. *Autism and Child Psychopathology Series*. Springer, Cham. https://doi.org/10.1007/978-3-319-61738-1_14

*Rahimi Pordanjani, S. (2021). Effectiveness of drama therapy on social skills of autistic children. *Practice in Clinical Psychology*, *9*(1), 9–18. https://doi.org/10.32598/jpcp.9.1.344.2

Del Re, A. (2013). *Compute.es: Compute effect sizes* (Version 0.2–2) [Computer software]. R package. https://www.acdelre.com

Del Re, A. C. (2015) A practical tutorial on conducting meta-analysis in R. *The Quantitative Methods for Psychology, 11*(1), 37–50. https://doi.org/10.20982/tqmp.11.1.p037

Del Re, A. C., & Hoyt, W. T. (2018). *MAd: Meta-analysis with mean differences* (Version 0.8) [Computer software]. R package. https://cran.rproject.org/web/packages/MAd/MAd.pdf

Ridout, S., & Eldevik, S. (2023). Measures used to assess treatment outcomes in children with autism receiving early and intensive behavioral interventions: A review. *Review Journal of Autism and Developmental Disorders*, 1–13. https://doi.org/10.1007/s40489-023-00355-9

RStudio Team. (2015). *RStudio: Integrated Development Environment for R* [Computer software]. http://www.rstudio.com/

Sansi, A., Nalbant, S., & Ozer, D. (2021). Effects of an inclusive physical activity program on the motor skills, social skills and attitudes of students with and without autism spectrum disorder. *Journal of Autism and Developmental Disorders, 51*(7), 2254–2270. https://doi.org/10.1007/s10803-020-04693-z

Schiltz, H., Gohari, D., Park, J., & Lord, C. (2024). A longitudinal study of loneliness in autism and otherneurodevelopmental disabilities: Coping with loneliness from childhood through adulthood. *Autism, 28*(6), 1471–1486. https://doi.org/10.1177/13623613231217337

Schohl, K. A., Van Hecke, A. V., Carson, A. M., Dolan, B., Karst, J., & Stevens, S. (2014). A replication and extension of the PEERS intervention: Examining effects on social skills and social anxiety in adolescents with autism spectrum disorders. *Journal of Autism and Developmental Disorders, 44*(3), 532–545. https://doi.org/10.1007/s10803-013-1900-1

Soares, E. E., Bausback, K., Beard, C. L., Higinbotham, M., Bunge, E. L., & Gengoux, G. W. (2021). Social skills training for autism spectrum disorder: A meta-analysis of in-person and technological interventions. *Journal of Technology in Behavioral Science, 6*, 166–180. https://doi.org/10.1007/s41347-020-00177-0

Steeger, C. M., Buckley, P. R., Pampel, F. C., Gust, C. J., & Hill, K. G. (2021). Common methodological problems in randomized controlled trials of preventive interventions. *Prevention Science, 22*(8), 1159–1172. https://doi.org/10.1007/s11121-021-01263-2

Symes, W., & Humphrey, N. (2010). Peer-group indicators of social inclusion among pupils with autistic spectrum disorders (ASD) in mainstream secondary schools. *School Psychology, 31*(5), 478–494. https://doi.org/10.1177/0143034310382496

Taylor, E. C., Livingston, L. A., Clutterbuck, R. A., Callan, M. J., & Shah, P. (2023). Psychological strengths andwell-being: Strengths use predicts quality of life, well-being and mental health in autism. *Autism, 27*(6), 1826–1839.https://doi.org/10.1177/13623613221146440

Thielmann, I., Zimmermann, J., Leising, D., & Hilbig, B. E. (2017). Seeing is knowing: On the predictive accuracy of self– and informant reports for prosocial and moral behaviours. *European Journal of Personality, 31*(4), 404–418. https://doi.org/10.1002/per.2112

Ting, V., & Weiss, J. A. (2017). Emotion regulation and parent co-regulation in children with autism spectrum disorder. *Journal of Autism and Developmental Disorders, 47*, 680–689. https://doi.org/10.1007/s10803-016-3009-9

Van Hecke, A. V., Stevens, S., Carson, A. M., Karst, J. S., Dolan, B., Schohl, K., McKindles, R. J., Remmel, R., & Brockman, S. (2015). Measuring the plasticity of social approach: A randomised controlled trial of the effects of the PEERS intervention on EEG asymmetry in adolescents with autism spectrum disorders. *Journal of Autism and Developmental Disorders, 45*(2), 316–335. https://doi.org/10.1007/s10803-013-1883-y

West, S.,L, Gartlehner, G., Mansfield, A.,J., et al. (2010). *Comparative Effectiveness Review Methods: Clinical Heterogeneity*. Agency for Healthcare Research and Quality (US).

Wolstencroft, J., Robinson, L., Srinivasan, R., Kerry, E., Mandy, W., & Skuse, D. (2018). A systematic review of group social skills interventions, and meta-analysis of outcomes, for children with high functioning ASD. *Journal of Autism and Developmental Disorders, 48*(7), 2293–2307. https://doi.org/10.1007/s10803-018-3485-1

Yizengaw, S. S. (2021). Effect of social skills training on interpersonal interactions of children with autism: An interventional research. *International Journal of Developmental Disabilities, 68*(6), 858–866. https://doi.org/10.1080/20473869.2021.1902730

Zisk, A. H., & Dalton, E. (2019). Augmentative and alternative communication for speaking autistic adults:Overview and recommendations. *Autism in Adulthood, 1*(2), 93–100. https://doi.org/10.1089/aut.2018.0007

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.